A Study on Topological Magnons

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Part I

Part I: An Introduction to Topological Matter

CHAPTER 1

MOTIVATIONS FOR TOPOLOGY IN STUDYING PHYSICS AND BAND STRUCTURES

The use of topology in condensed matter physics arises from the realization that certain electronic properties of materials cannot be characterized solely by local order parameters or symmetry breaking, as in traditional phases of matter. Instead, topological concepts classify phases based on global properties of the electronic wavefunctions, such as Berry phases, Berry curvature, and associated topological invariants like Chern numbers. In band theory, this allows one to distinguish between phases that have identical energy spectra but fundamentally different electronic structures, leading to robust physical phenomena such as quantized Hall conductance and protected edge states. These features are immune to local perturbations as long as the bulk energy gap remains open, making topology a powerful framework for understanding and predicting novel quantum materials, including topological insulators, semimetals, and superconductors.

1 Electronic Band Structures and Brillouin Zones

In crystalline solids, electrons experience a periodic potential due to the underlying lattice. According to Bloch's theorem, electron wavefunctions take the form

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}),$$

where $u_{n\mathbf{k}}(\mathbf{r})$ has the periodicity of the lattice, n is the band index, and $\mathbf{k}$ is the crystal momentum. The eigenenergies $E_n(\mathbf{k})$ form the *electronic band structure* of the material.

The reciprocal lattice vectors $\mathbf{G}$ define the *Brillouin zone* (BZ), the primitive cell in reciprocal space. The first BZ contains all unique crystal momenta and is central in understanding band dispersions. Band gaps arise when Bragg scattering at zone boundaries leads to avoided crossings in $E(\mathbf{k})$.

2 Su–Schrieffer–Heeger (SSH) Model

2.1 Introduction and Physical Motivation

The Su–Schrieffer–Heeger (SSH) model was originally introduced to describe the electronic structure of polyacetylene, a quasi-one-dimensional polymer with alternating single and double bonds [6]. The SSH model captures essential features of *dimerized* one-dimensional lattices, highlighting how lattice symmetry breaking can open an electronic band gap and lead to topologically distinct insulating phases.

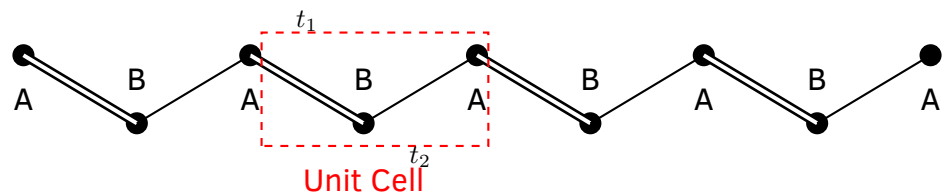

Figure 1.1: Unit cell of the polyacetylene-like SSH model with staggered sites (A/B) and alternating bond strengths. The dashed red box highlights one unit cell, where t_1 (double bond) is the strong intra-cell hopping and t_2 (single bond) is the weak inter-cell hopping. The chain continues periodically in both directions.

2.2 Lattice and Tight-Binding Assumptions

We consider a one-dimensional chain composed of unit cells, each containing two inequivalent atoms (labeled A and B). The hopping amplitudes between neighboring sites alternate:

- t_1: hopping amplitude within a unit cell (intracell).
- t_2: hopping amplitude between neighboring cells (intercell).

We make the following assumptions:

- The electrons are described within a *tight-binding approximation*, where we consider only nearest-neighbor hopping.
- Spin degrees of freedom are ignored (focus on spinless fermions).
- Overlap between atomic orbitals beyond nearest neighbors is negligible.

2.3 Real-Space Hamiltonian

The Su–Schrieffer–Heeger (SSH) Hamiltonian describes electrons hopping on a one-dimensional bipartite lattice with alternating hopping amplitudes:

$$\hat{H} = t_1 \sum_n \left(|A_n\rangle\langle B_n| + |B_n\rangle\langle A_n| \right) + t_2 \sum_n \left(|A_{n+1}\rangle\langle B_n| + |B_n\rangle\langle A_{n+1}| \right). \tag{1.1}$$

Here, $|A_n\rangle$ and $|B_n\rangle$ are orthonormal Wannier states localized at the two inequivalent sites of unit cell n, and t_1, t_2 are intracell and intercell hopping amplitudes, respectively.

2.4 Fourier Transform to Momentum Space

We define Bloch basis states via:

$$|A_k\rangle = \frac{1}{\sqrt{N}} \sum_n e^{-ikn} |A_n\rangle, \qquad |B_k\rangle = \frac{1}{\sqrt{N}} \sum_n e^{-ikn} |B_n\rangle, \tag{1.2}$$

where $k \in [-\pi, \pi]$ and N is the number of unit cells.

Substituting into Eq. (1.1), the Hamiltonian becomes:

$$\hat{H} = \sum_k \Big[(t_1 + t_2 e^{-ik}) |A_k\rangle\langle B_k| + (t_1 + t_2 e^{ik}) |B_k\rangle\langle A_k| \Big]. \tag{1.3}$$

2.5 Eigenvalue Problem

We look for eigenstates in the form:

$$|\psi_k\rangle = u_k |A_k\rangle + v_k |B_k\rangle, \tag{1.4}$$

such that:

$$\hat{H}|\psi_k\rangle = E(k)|\psi_k\rangle. \tag{1.5}$$

Operating with Eq. (1.3) gives:

$$(t_1 + t_2 e^{-ik}) v_k = E(k) u_k, \tag{1.6}$$

$$(t_1 + t_2 e^{ik}) u_k = E(k) v_k. \tag{1.7}$$

Multiplying yields the characteristic equation:

$$E^2(k) = (t_1 + t_2 e^{-ik})(t_1 + t_2 e^{ik}) = t_1^2 + t_2^2 + 2t_1 t_2 \cos k. \tag{1.8}$$

Thus, the eigenvalues (dispersion relations) are:

$$E_\pm(k) = \pm\sqrt{t_1^2 + t_2^2 + 2t_1 t_2 \cos k}. \tag{1.9}$$

2.6 Eigenvectors

From the eigenvalue equation:

$$v_k = \frac{E(k)}{t_1 + t_2 e^{-ik}} u_k, \tag{1.10}$$

choosing normalization gives eigenstates:

$$|u_\pm(k)\rangle = \frac{1}{\sqrt{2}} \left(e^{-i\phi(k)} |A_k\rangle \pm |B_k\rangle \right), \tag{1.11}$$

where the phase of hopping is:

$$\phi(k) = \arg(t_1 + t_2 e^{-ik}). \tag{1.12}$$

2.7 Summary of Bra–Ket Formulation

The SSH Hamiltonian describes hopping between A and B sites using Eq. (1.1). - Fourier transforming leads to a k-space Hamiltonian (Eq. (1.3)) with momentum-dependent complex hopping. - Solving yields two symmetric energy bands separated by a gap:

$$\Delta = 2|t_1 - t_2|. \tag{1.13}$$

The eigenvectors acquire a nontrivial momentum-dependent phase $\phi(k)$, which plays a crucial role in Berry phase and topological properties.

2.8 Physical Picture of Band Formation

In the absence of dimerization ($t_1 = t_2$), the SSH model reduces to a uniform tight-binding chain with metallic behavior (gapless at $k = \pi$). Introducing dimerization breaks inversion symmetry between bonds and opens a band gap, turning the system into an insulator. Depending on whether $t_1 > t_2$ or $t_1 < t_2$, the resulting insulating phase is topologically trivial or nontrivial, respectively, leading to edge states in finite chains.

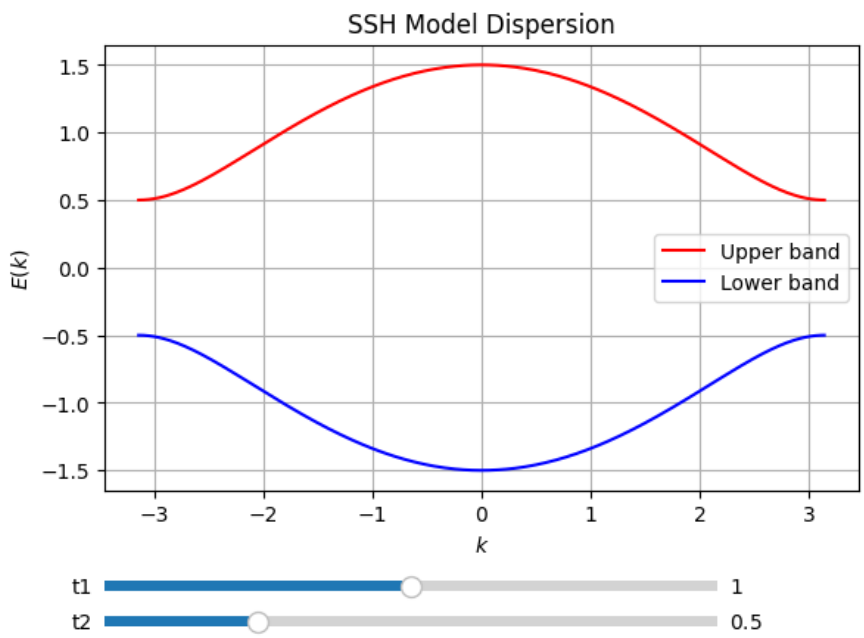

Figure 1.2: Dispersion relation of the SSH model with parameters $t_1 = 1.0$, $t_2 = 0.5$, and $a = 1.0$. A band gap $2|t_1 - t_2|$ separates the two bands.

2.9 Summary

The SSH model, with only two parameters (t_1 and t_2), illustrates how topology can emerge in simple tight-binding systems. The resulting dispersion relation (Eq.1.9) shows two symmetric bands separated by a tunable gap, whose closing and reopening signify a change in the topological invariant (the Zak phase).

CHAPTER 2

TOPOLOGICAL QUANTUM MATTER BASICS

1 Berry Phase, Quantum Holonomy, and the Topology of the Aharonov–Bohm Effect

1.1 Quantum Holonomy and Geometric Phase

In classical differential geometry, *holonomy* refers to the transformation experienced by a vector when it is parallel transported around a closed loop in a curved space. If the space has curvature or nontrivial topology, the transported vector generally fails to return to its original orientation, acquiring a geometrical change determined by the underlying connection.

In quantum mechanics, the state of a system is described by a ray in Hilbert space, i.e., a complex vector modulo an overall phase. Given a parameter-dependent Hamiltonian $H(\boldsymbol{R})$, with instantaneous eigenstates $|n(\boldsymbol{R})\rangle$, we consider adiabatic transport of a quantum state along a closed loop $\mathcal{C}$ in parameter space. Because global phase has no physical meaning, only relative phases matter. Imposing the *parallel transport condition*:

$$\langle n(\boldsymbol{R})|\dot{n}(\boldsymbol{R})\rangle = 0, \tag{2.1}$$

ensures that the state accumulates no dynamical phase along an infinitesimal segment. After traversing $\mathcal{C}$, however, the final state generally differs by a phase:

$$|n(\boldsymbol{R}(T))\rangle = e^{i\gamma_n[\mathcal{C}]}\,|n(\boldsymbol{R}(0))\rangle\,. \tag{2.2}$$

This phase,

$$\gamma_n[\mathcal{C}] = i\oint_{\mathcal{C}} \langle n(\boldsymbol{R})|\nabla_{\boldsymbol{R}} n(\boldsymbol{R})\rangle \cdot d\boldsymbol{R}, \tag{2.3}$$

is the *Berry phase*, which is purely geometric because it depends only on the shape of the path in parameter space and not on the rate of traversal.

1.2 Quantum Holonomy as a Gauge Connection

The quantity:

$$\mathbf{A}_n(\boldsymbol{R}) = i\,\langle n(\boldsymbol{R})|\nabla_{\boldsymbol{R}} n(\boldsymbol{R})\rangle\,, \tag{2.4}$$

is the *Berry connection*, analogous to a gauge potential in electromagnetism. Its curl defines the *Berry curvature*:

$$\boldsymbol{\Omega}_n(\boldsymbol{R}) = \nabla_{\boldsymbol{R}} \times \mathbf{A}_n(\boldsymbol{R}), \tag{2.5}$$

which plays the role of a fictitious magnetic field in parameter space. The Berry phase is then identified with the holonomy of the Berry connection:

$$\gamma_n[\mathcal{C}] = \oint_{\mathcal{C}} \mathbf{A}_n \cdot d\boldsymbol{R} = \int_{S(\mathcal{C})} \boldsymbol{\Omega}_n \cdot d\mathbf{S}, \tag{2.6}$$

where $S(\mathcal{C})$ is any surface bounded by $\mathcal{C}$. Thus, the Berry phase is the quantum analog of holonomy in fiber bundles, with the eigenstates forming a $U(1)$ fiber over parameter space.

1.3 Geometric Phase and Topology

When parameter space is simply connected and curvature is zero, the Berry phase around any contractible loop vanishes. However, if parameter space has nontrivial topology (e.g., missing points, singularities, or periodic boundary conditions), then loops that are topologically nontrivial can acquire nonzero geometric phases even in the absence of local curvature. This is the origin of many topological phenomena in quantum mechanics, where phase information is encoded in global, rather than local, properties.

1.4 Topological Origin of the Aharonov–Bohm Effect

A paradigmatic example is the *Aharonov–Bohm (AB) effect*. Consider a charged particle moving in a region where the magnetic field $\mathbf{B} = \nabla \times \mathbf{A}$ vanishes everywhere accessible to the particle, but the vector potential $\mathbf{A}$ is not globally a pure gauge due to a confined flux Φ_B in an excluded region (e.g., inside an ideal solenoid). Even though $\mathbf{B} = 0$ along the path, the wavefunction acquires a phase:

$$\gamma_{\mathrm{AB}} = \frac{q}{\hbar} \oint_{\mathcal{C}} \mathbf{A} \cdot d\mathbf{r} = \frac{q\Phi_B}{\hbar}. \tag{2.7}$$

This phase shift has observable consequences, such as shifts in interference fringes, and is independent of the local electromagnetic field along the path.

The AB phase is topological because:

1. It depends only on the winding number of the path around the flux tube.

2. It cannot be removed by any smooth local gauge transformation, since the region inside the solenoid is excluded from the configuration space, making it multiply connected.

This directly illustrates how holonomy and geometric phases in quantum mechanics arise naturally from topological features of parameter or configuration space.

1.5 Summary

The Berry phase is a manifestation of quantum holonomy in parameter space, arising from parallel transport of quantum states and described mathematically using gauge connections and curvature. When topology enters, as in the Aharonov–Bohm effect, these geometric phases become robust and globally quantized, playing a fundamental role in modern topological phases of matter.

2 Quantum Hall Effect and Its Topological Origin

2.1 Introduction

The Hall effect, first discovered by Edwin Hall in 1879, refers to the development of a transverse voltage V_H across a conducting sample when an electric current flows in the presence of a perpendicular magnetic field B. The Hall resistance is defined as:

$$R_H = \frac{V_H}{I} = \frac{B}{ne}, \tag{2.8}$$

where n is the carrier density and e is the electron charge.

In 1980, Klaus von Klitzing and co-workers discovered that at very low temperatures ($T \lesssim 4\,\mathrm{K}$) and high magnetic fields, the Hall resistance of a two-dimensional electron gas (2DEG) confined in a semiconductor heterostructure becomes quantized [7]:

$$R_H = \frac{h}{e^2}\frac{1}{\nu}, \tag{2.9}$$

where ν is an integer known as the *filling factor*. This phenomenon is known as the *Integer Quantum Hall Effect* (IQHE).

The longitudinal resistance R_{xx} vanishes in the same regime, indicating dissipationless transport:

$$R_{xx} = 0 \quad \text{and} \quad R_H = \frac{h}{e^2\nu}. \tag{2.10}$$

The quantization in Eq. (2.9) is universal and insensitive to sample geometry, disorder, or material details, depending only on fundamental constants.

2.2 Landau Levels and the Origin of Quantization

A 2DEG in a perpendicular magnetic field experiences quantization of electron motion into discrete energy levels called Landau levels:

$$E_n = \hbar\omega_c\left(n + \frac{1}{2}\right), \quad \omega_c = \frac{eB}{m^*}, \tag{2.11}$$

where m^* is the effective mass and ω_c is the cyclotron frequency.

Each Landau level has a degeneracy proportional to the magnetic flux through the system:

$$N_\phi = \frac{BA}{\Phi_0}, \quad \Phi_0 = \frac{h}{e}. \tag{2.12}$$

When an integer number ν of Landau levels are completely filled, the Fermi level lies in a mobility gap and backscattering is suppressed, resulting in the quantized Hall conductivity:

$$\sigma_{xy} = \nu \frac{e^2}{h}, \qquad \sigma_{xx} = 0. \tag{2.13}$$

2.3 Topological Interpretation: Chern Number

The robustness of the IQHE can be understood from topology. In a periodic crystal under a magnetic field (or using the Peierls substitution), the electronic wavefunctions acquire a geometric phase known as the Berry phase [9]. The Berry curvature in momentum space for a Bloch state $|u_n(\mathbf{k})\rangle$ of the n-th band is:

$$\Omega_n(\mathbf{k}) = i \left(\left\langle \frac{\partial u_n}{\partial k_x} \middle| \frac{\partial u_n}{\partial k_y} \right\rangle - \left\langle \frac{\partial u_n}{\partial k_y} \middle| \frac{\partial u_n}{\partial k_x} \right\rangle \right). \tag{2.14}$$

Integrating the Berry curvature over the Brillouin zone (BZ) gives an integer known as the *Chern number*:

$$C_n = \frac{1}{2\pi} \int_{\mathrm{BZ}} \Omega_n(\mathbf{k})\, d^2k \in \mathbb{Z}. \tag{2.15}$$

The Hall conductivity is then expressed as:

$$\sigma_{xy} = \frac{e^2}{h} \sum_{n \in \text{filled}} C_n. \tag{2.16}$$

Hence, the quantization of Hall conductance is a direct manifestation of a topological invariant (the Chern number) of the electronic band structure [8]. This topological perspective explains the extraordinary robustness of the IQHE to disorder and microscopic imperfections.

2.4 Edge States and Bulk–Boundary Correspondence

The IQHE also features chiral edge states, which propagate along the boundaries of the sample. These edge states are protected by topology and are immune to backscattering from non-magnetic impurities. The number of chiral edge modes is equal to the total Chern number of the filled bulk bands:

$$N_{\text{edge}} = \sum_{n \in \text{filled}} C_n. \tag{2.17}$$

This is an example of the *bulk–boundary correspondence*, a central concept in topological phases of matter.

2.5 Summary

- The Integer Quantum Hall Effect arises from the quantization of electron motion into Landau levels and is characterized by vanishing longitudinal resistance and quantized Hall resistance.

- The quantized Hall conductance is a topological invariant given by the sum of Chern numbers of filled bands.
- Robust chiral edge states emerge, protected by the topological properties of the bulk.

3 Berry Phases, Chern Numbers, and Topology in Quantum Hall Systems

3.1 Berry Phase and Adiabatic Evolution

Consider a quantum system described by a Hamiltonian $\hat{H}(\mathbf{R})$ that depends smoothly on a set of external parameters $\mathbf{R}$. For an instantaneous eigenstate $\hat{H}(\mathbf{R})|n(\mathbf{R})\rangle = E_n(\mathbf{R})|n(\mathbf{R})\rangle$, the adiabatic theorem states that under slow evolution of $\mathbf{R}(t)$ along a closed path $\mathcal{C}$ in parameter space, the wavefunction acquires a total phase:

$$\gamma_n(\mathcal{C}) = i \oint_{\mathcal{C}} \langle n(\mathbf{R})|\nabla_{\mathbf{R}} n(\mathbf{R})\rangle \cdot d\mathbf{R}, \tag{2.18}$$

known as the *Berry phase* [9]. It is purely geometric and gauge-invariant modulo 2π.

3.2 Berry Connection and Curvature

Defining the *Berry connection*:

$$\mathbf{A}_n(\mathbf{R}) = i\langle n(\mathbf{R})|\nabla_{\mathbf{R}} n(\mathbf{R})\rangle, \tag{2.19}$$

the Berry curvature is its curl:

$$\mathbf{\Omega}_n(\mathbf{R}) = \nabla_{\mathbf{R}} \times \mathbf{A}_n(\mathbf{R}). \tag{2.20}$$

In two dimensions (R_1, R_2), the Berry curvature reduces to:

$$\Omega_n(R_1, R_2) = \frac{\partial A_{2,n}}{\partial R_1} - \frac{\partial A_{1,n}}{\partial R_2}. \tag{2.21}$$

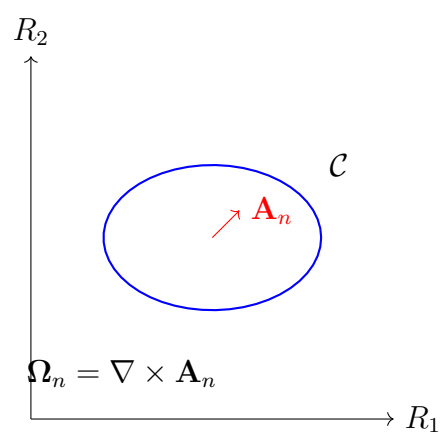

Figure 2.1: Illustration of a closed path $\mathcal{C}$ in parameter space leading to a Berry phase.

3.3 Chern Number and Gauss–Bonnet Analogy

The integral of Berry curvature over a compact two-dimensional manifold M defines the *first Chern number*:

$$C_n = \frac{1}{2\pi} \int_M \Omega_n(R_1, R_2)\, dR_1 dR_2, \tag{2.22}$$

which is always an integer. This integer is topologically invariant and robust under continuous deformations that do not close the energy gap.

This is analogous to the Gauss–Bonnet theorem in differential geometry:

$$\int_M K\, dA = 2\pi\chi(M), \tag{2.23}$$

where K is the Gaussian curvature and $\chi(M)$ is the Euler characteristic. Here, Berry curvature plays a role analogous to Gaussian curvature, and the Chern number plays a role analogous to the Euler characteristic.

3.4 Topology of Brillouin Zones

For a periodic crystal, the momentum space is defined modulo reciprocal lattice vectors, forming a compact manifold equivalent to a two-dimensional torus T^2. Bloch eigenstates $|u_n(\mathbf{k})\rangle$ define a fiber bundle over this torus. The Berry curvature on the Brillouin zone torus gives rise to a Chern number:

$$C_n = \frac{1}{2\pi} \int_{\mathrm{BZ}} \Omega_n(\mathbf{k})\, d^2k \in \mathbb{Z}. \tag{2.24}$$

This integer classifies the topology of the electronic band.

3.5 Chern Theorem for Quantum Hall Conductance

The Thouless–Kohmoto–Nightingale–den Nijs (TKNN) theorem [8] relates the Hall conductance of a filled band to its Chern number:

$$\sigma_{xy} = \frac{e^2}{h} \sum_{n \in \mathrm{filled}} C_n. \tag{2.25}$$

Proof Sketch: (1) Express σ_{xy} via the Kubo formula. (2) Show that it can be written as an integral of Berry curvature over the Brillouin zone. (3) Use the fact that the integral of Berry curvature is quantized as in Eq. (2.22).

3.6 Hofstadter Butterfly: Lattice + Magnetic Field

For electrons on a 2D square lattice with lattice constant a in a uniform magnetic field, the energy spectrum as a function of magnetic flux per plaquette ϕ/ϕ_0 exhibits a fractal structure known as the Hofstadter butterfly:

$$\phi = Ba^2, \qquad \phi_0 = \frac{h}{e}. \tag{2.26}$$

Each band gap is associated with a quantized Hall conductivity determined by a Chern number. This model explicitly demonstrates how topology and magnetic flux conspire to produce intricate band structures.

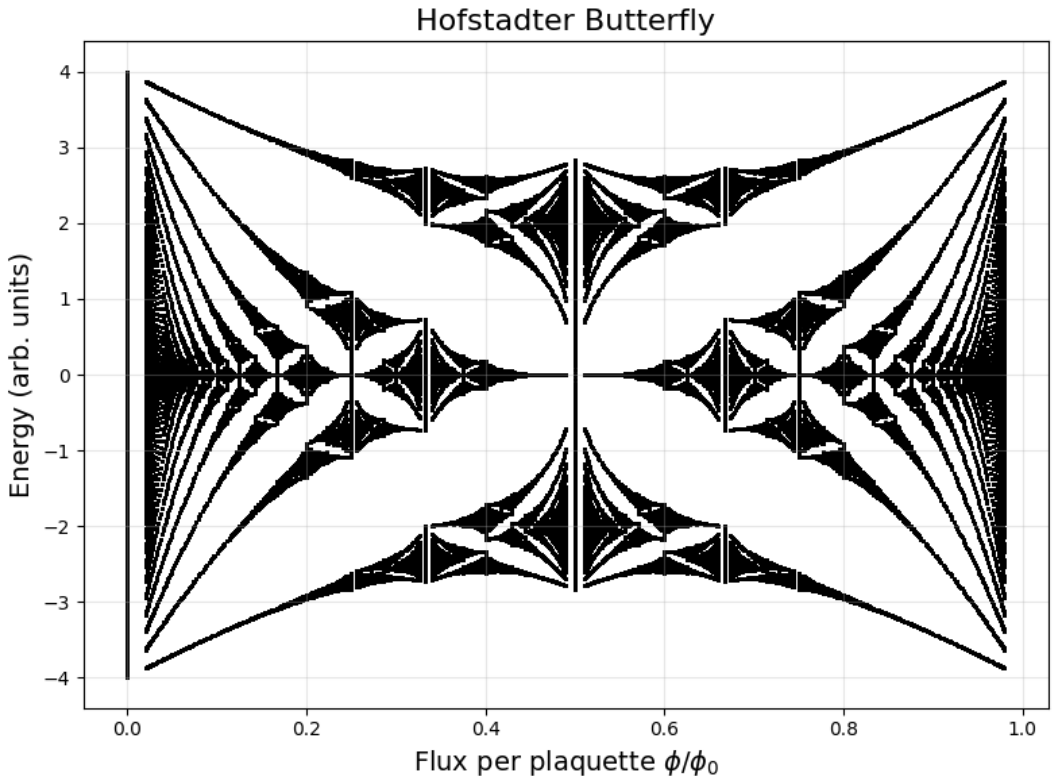

Figure 2.2: Hofstadter butterfly spectrum for electrons on a square lattice with magnetic flux per plaquette ϕ/ϕ_0. Each gap is associated with an integer Chern number. (Referred from Symmetry, Broken Symmetry, and Topology in Modern Physics: A First Course [13])

3.7 Summary

- Berry phases arise from geometric properties of parameter space.
- Berry curvature acts like a magnetic field in parameter space, and its integral (Chern number) is topologically quantized.
- Brillouin zones have toroidal topology, making Chern numbers natural invariants for band structures.
- The IQHE Hall conductivity is given by a sum of band Chern numbers.
- The Hofstadter butterfly provides a lattice realization where topology and magnetic flux produce fractal spectra.

Part II

Part II: What are Magnons?

CHAPTER 1

THEORY BEHIND MAGNONS

1 Theory of Spin Waves and Magnon Dispersion Relations

1.1 Heisenberg Model and Spin-Wave Excitations

A microscopic model for magnetic interactions is provided by the Heisenberg Hamiltonian:

$$\hat{H} = -J \sum_{\langle i,j \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j - g\mu_B B \sum_i \hat{S}_i^z, \tag{1.1}$$

where J is the exchange interaction constant, $\hat{\mathbf{S}}_i = (\hat{S}_i^x, \hat{S}_i^y, \hat{S}_i^z)$ are spin operators at lattice site i, g is the Landé g-factor, μ_B is the Bohr magneton, and B is an applied magnetic field. For ferromagnetic interactions ($J > 0$), the ground state corresponds to all spins aligned. Small deviations from this fully aligned state give rise to collective excitations known as *spin waves* [1].

1.2 Holstein–Primakoff Bosonization

To handle quantum fluctuations around the ordered ground state, we use bosonization, mapping spin operators to bosonic creation and annihilation operators $\hat{a}_i^\dagger$ and $\hat{a}_i$. The Holstein–Primakoff (HP) transformation [2] is:

$$\hat{S}_i^+ = \sqrt{2S - \hat{a}_i^\dagger \hat{a}_i}\, \hat{a}_i, \qquad \hat{S}_i^- = \hat{a}_i^\dagger \sqrt{2S - \hat{a}_i^\dagger \hat{a}_i}, \qquad \hat{S}_i^z = S - \hat{a}_i^\dagger \hat{a}_i, \tag{1.2}$$

where S is the spin quantum number. For low-lying excitations ($\langle \hat{a}_i^\dagger \hat{a}_i \rangle \ll 2S$), we expand:

$$\hat{S}_i^+ \approx \sqrt{2S}\, \hat{a}_i, \qquad \hat{S}_i^- \approx \sqrt{2S}\, \hat{a}_i^\dagger. \tag{1.3}$$

Substituting Eq. (1.2) into Eq. (1.1) and keeping quadratic terms yields:

$$\begin{aligned} \hat{H} &= -J \sum_{\langle i,j \rangle} \left[(S - \hat{a}_i^\dagger \hat{a}_i)(S - \hat{a}_j^\dagger \hat{a}_j) + \frac{1}{2} \left(\hat{S}_i^+ \hat{S}_j^- + \hat{S}_i^- \hat{S}_j^+ \right) \right] - g\mu_B B \sum_i (S - \hat{a}_i^\dagger \hat{a}_i) \\ &= E_0 + JS \sum_{\langle i,j \rangle} \left(\hat{a}_i^\dagger \hat{a}_i + \hat{a}_j^\dagger \hat{a}_j - \hat{a}_i^\dagger \hat{a}_j - \hat{a}_j^\dagger \hat{a}_i \right) + g\mu_B B \sum_i \hat{a}_i^\dagger \hat{a}_i, \end{aligned} \tag{1.4}$$

where $E_0 = -zNJS^2 - g\mu_B BNS$ is the classical ground state energy (z is the number of nearest neighbors, N the number of sites).

1.3 Fourier Transform and Diagonalization

To exploit translational invariance, we define:

$$\hat{a}_i = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} \hat{a}_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{R}_i}. \tag{1.5}$$

Substituting into Eq. (1.4) gives:

$$\hat{H} = E_0 + \sum_{\mathbf{k}} \left[g\mu_B B + 2JSz(1-\gamma_{\mathbf{k}}) \right] \hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}}, \tag{1.6}$$

where:

$$\gamma_{\mathbf{k}} = \frac{1}{z} \sum_{\boldsymbol{\delta}} e^{i\mathbf{k}\cdot\boldsymbol{\delta}}, \tag{1.7}$$

and $\boldsymbol{\delta}$ runs over nearest-neighbor vectors. Hence, the eigenmodes are bosonic quasiparticles called *magnons*, with dispersion:

$$\boxed{\hbar\omega(\mathbf{k}) = g\mu_B B + 2JSz\Big(1-\gamma_{\mathbf{k}}\Big)}. \tag{1.8}$$

For a one-dimensional chain with lattice spacing a, $z = 2$, and $\gamma_k = \cos(ka)$:

$$\hbar\omega(k) = g\mu_B B + 4JS\sin^2\left(\frac{ka}{2}\right). \tag{1.9}$$

1.4 Antiferromagnetic Case

For an antiferromagnet, we separate spins into two sublattices and perform a Bogoliubov transformation. The resulting dispersion relation is:

$$\boxed{\hbar\omega_{\mathrm{AF}}(k) = 2JSz\sqrt{1-\gamma_k^2}} \tag{1.10}$$

which is linear for small k, in contrast to the quadratic behavior of ferromagnets [3, 10].

1.5 Low-Wavelength Limit

For small wavevectors ($ka \ll 1$),

$$\cos(ka) \approx 1 - \frac{k^2a^2}{2}, \tag{1.11}$$

giving:

$$\hbar\omega(k) \approx g\mu_B B + Dk^2, \qquad D = JSa^2z, \tag{1.12}$$

where D is the spin-wave stiffness constant. This quadratic dependence is characteristic of ferromagnetic magnons, while antiferromagnetic magnons exhibit a linear acoustic-like spectrum.

1.6 Figure of the Dispersion Relations

Figure 1.3 shows the calculated dispersion relations for ferromagnetic and antiferromagnetic systems based on Eqs. (1.8) and (1.10).

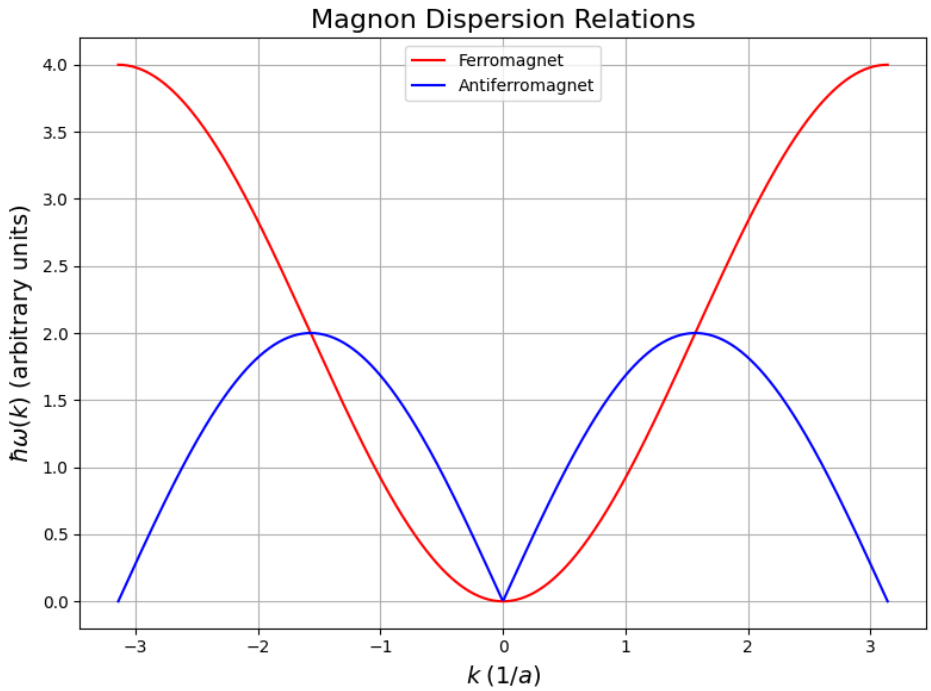

Figure 1.1: Dispersion relation of magnons for ferromagnetic (red) and antiferromagnetic (blue) spin waves. The ferromagnetic case shows a quadratic low-k behavior, while the antiferromagnetic case is linear near $k = 0$. (Figure generated using Python code described in the text.)

1.7 Physical Interpretation

Spin waves represent collective precessions of spins about their equilibrium direction. Through bosonization, these collective excitations are described as non-interacting bosons known as *magnons*. Each magnon reduces the total magnetization by one unit of $\hbar$ and carries energy $\hbar\omega(k)$, determining thermal properties and dynamic responses of magnetic materials.

CHAPTER 2

HOW DOES TOPOLOGY COME IN TO PLAY?

1 Topological Magnons in Systems with Heisenberg and DM Interactions

1.1 Heisenberg Exchange Hamiltonian

The starting point for most magnetic systems is the Heisenberg Hamiltonian:

$$\hat{H}_{\text{Heis}} = -J \sum_{\langle i,j \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j - g\mu_B \mathbf{B} \cdot \sum_i \hat{\mathbf{S}}_i, \tag{2.1}$$

where J is the nearest-neighbor exchange coupling, g the Landé g-factor, and $\mathbf{B}$ the applied magnetic field. For $J > 0$ the ground state is ferromagnetic, while for $J < 0$ it is antiferromagnetic.

1.2 Dzyaloshinskii–Moriya Interaction (DMI)

In crystals lacking inversion symmetry, spin-orbit coupling induces an additional term known as the Dzyaloshinskii–Moriya interaction (DMI) [4, 5]:

$$\hat{H}_{\text{DMI}} = \sum_{\langle i,j \rangle} \mathbf{D}_{ij} \cdot (\hat{\mathbf{S}}_i \times \hat{\mathbf{S}}_j), \tag{2.2}$$

where $\mathbf{D}_{ij}$ is the DMI vector determined by crystal symmetry. This term favors canted spin textures and breaks inversion symmetry in spin space, introducing a preferred chirality to spin configurations.

1.3 Combined Hamiltonian and Ground State

The total Hamiltonian is:

$$\hat{H} = -J \sum_{\langle i,j \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \sum_{\langle i,j \rangle} \mathbf{D}_{ij} \cdot (\hat{\mathbf{S}}_i \times \hat{\mathbf{S}}_j) - g\mu_B B \sum_i \hat{S}_i^z. \tag{2.3}$$

For a uniform ferromagnet ($J > 0$) in a sufficiently strong B, the spins align along z.

1.4 Bosonization via Holstein–Primakoff Transformation

We now map spins onto bosons using the Holstein–Primakoff representation:

$$\hat{S}_i^z = S - \hat{a}_i^\dagger \hat{a}_i, \qquad \hat{S}_i^+ = \sqrt{2S}\,\hat{a}_i, \qquad \hat{S}_i^- = \sqrt{2S}\,\hat{a}_i^\dagger. \tag{2.4}$$

Substituting into Eq. (2.3) and keeping only quadratic terms in bosons (linear spin-wave approximation) gives:

$$\hat{H} = E_0 + \sum_{\mathbf{k}} \hat{\Psi}_{\mathbf{k}}^\dagger \mathcal{H}(\mathbf{k}) \hat{\Psi}_{\mathbf{k}}, \quad \hat{\Psi}_{\mathbf{k}} = \begin{pmatrix} \hat{a}_{\mathbf{k}} \\ \hat{a}_{-\mathbf{k}}^\dagger \end{pmatrix}. \tag{2.5}$$

1.5 Spin-Wave Hamiltonian with DMI

For a 2D honeycomb ferromagnet with out-of-plane DMI ($\mathbf{D}_{ij} = D\hat{z}$) and uniform ferromagnetic order:

$$\mathcal{H}(\mathbf{k}) = \begin{pmatrix} A(\mathbf{k}) & B(\mathbf{k}) \\ B^*(\mathbf{k}) & A(-\mathbf{k}) \end{pmatrix}, \tag{2.6}$$

where:

$$A(\mathbf{k}) = g\mu_B B + JSz(1 - \gamma(\mathbf{k})), \tag{2.7}$$

$$B(\mathbf{k}) = -JS\gamma(\mathbf{k}) + iDS\xi(\mathbf{k}), \tag{2.8}$$

and $\gamma(\mathbf{k}), \xi(\mathbf{k})$ are geometric factors depending on lattice type. The DMI term enters as an imaginary hopping, similar to a complex phase in the Haldane model.

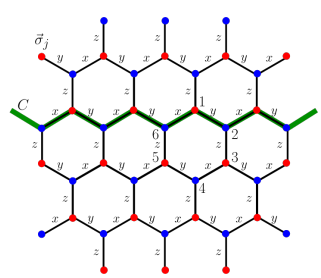

Figure 2.1: A honeycomb lattice and three types of bonds. On each vertex there is an $S = 1/2$ degree of freedom indicated by a Pauli matrix σ_j . C represents an arbitrary contour drawn on the lattice.

1.6 Topological Magnon Bands

Diagonalizing $\mathcal{H}(\mathbf{k})$ yields two magnon bands:

$$\varepsilon_\pm(\mathbf{k}) = A(\mathbf{k}) \pm \sqrt{|B(\mathbf{k})|^2}. \tag{2.9}$$

Due to the imaginary DMI contribution, a gap opens at Dirac points (K, K'), producing magnon bands analogous to topological electronic bands.

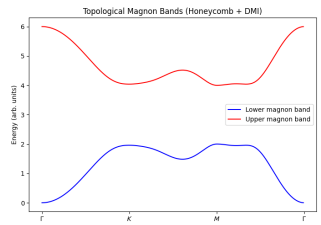

Figure 2.2: Plot of Dispersion Relation of Magnon Bands in the Honeycomb Lattice with $J = 1$, $D = 0.2$, $S = 1$, $a = 1$

1.7 Berry Curvature and Chern Number

The eigenvectors $|u_n(\mathbf{k})\rangle$ of $\mathcal{H}(\mathbf{k})$ define a Berry connection:

$$\mathbf{A}_n(\mathbf{k}) = i\langle u_n(\mathbf{k})|\nabla_{\mathbf{k}}|u_n(\mathbf{k})\rangle, \tag{2.10}$$

and Berry curvature:

$$\Omega_n(\mathbf{k}) = \nabla_{\mathbf{k}} \times \mathbf{A}_n(\mathbf{k}) \cdot \hat{z}. \tag{2.11}$$

Integration over the Brillouin zone gives the Chern number:

$$C_n = \frac{1}{2\pi} \int_{\mathrm{BZ}} \Omega_n(\mathbf{k})\, d^2k. \tag{2.12}$$

A nonzero C_n indicates topologically protected chiral edge magnons.

1.8 Physical Consequences

- **Magnon Hall Effect:** Due to nonzero Berry curvature, magnons experience a transverse deflection, leading to a thermal Hall effect observable experimentally [11].
- **Chiral Edge States:** Bulk-edge correspondence ensures that interfaces host unidirectional magnon edge modes robust against disorder.

Part III

Part III: Role of Inter-layer Spin Canting in Layered AFM: A Study

CHAPTER 1

TOPOLOGICAL DIRAC NODAL LINE MAGNONS IN LAYERED ANTIFERROMAGNETS

This chapter provides a comprehensive theoretical analysis of the symmetry-protected topological Dirac Nodal Line (DNL) magnonic phase in layered honeycomb collinear antiferromagnets, based on the foundational model by S. A. Owerre [12]. We provide an explicit derivation of the magnon dispersion relations, define the topological invariants, and analyze the effect of symmetry-breaking perturbations.

1 The Spin Hamiltonian Model

We begin with the microscopic spin Hamiltonian for a layered honeycomb antiferromagnet in the absence of an external magnetic field ($\mathbf{H} = 0$). The model includes intralayer and interlayer exchange interactions, and the Dzyaloshinskii-Moriya (DM) interaction.

1.1 The Next-Nearest-Neighbor DMI Interaction

The Dzyaloshinskii-Moriya interaction (DMI) is an antisymmetric exchange interaction that arises from spin-orbit coupling in crystal structures lacking a center of inversion symmetry. Unlike the standard Heisenberg exchange which favors collinear spin alignments, the DMI energetically favors a canting (tilting) of neighboring spins.

Mathematical Formulation

The general form of the DMI Hamiltonian is:

$$\mathcal{H}_{DM} = \sum_{\langle\langle ij \rangle\rangle} \mathbf{D}_{ij} \cdot (\mathbf{S}_i \times \mathbf{S}_j) \tag{1.1}$$

where the sum is over next-nearest-neighbor (NNN) pairs of spins, $\mathbf{S}_i$ and $\mathbf{S}_j$. The vector $\mathbf{D}_{ij}$ is the DM vector, whose direction and magnitude are determined by the crystal's symmetry.

For the honeycomb lattice, there is no inversion center between two NNN sites (which are on the same sublattice). This broken symmetry allows for a DM vector that is perpendicular to the honeycomb plane. Therefore, we can write:

$$\mathbf{D}_{ij} = D\nu_{ij}\hat{z} \tag{1.2}$$

Here, D is the magnitude of the DMI strength, and $\nu_{ij} = \pm 1$ is a sign factor that depends on the orientation of the bond connecting the NNN sites, as illustrated in Figure 1.1.

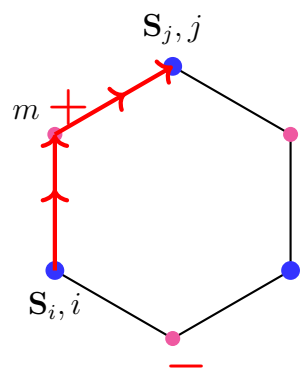

Figure 1.1: The NNN DMI path between two sites i and j on the same sublattice (blue). The sign ν_{ij} is determined by the intermediate site m (magenta). For a counter-clockwise path $i \to m \to j$, $\nu_{ij} = +1$, while for a clockwise path, it would be -1.

1.2 Model Hamiltonian

The full Hamiltonian, including the intralayer DMI, is described by [12]:

$$\mathcal{H} = J \sum_{\langle ij \rangle, l} \mathbf{S}_{i,l} \cdot \mathbf{S}_{j,l} + \sum_{\langle\langle ij \rangle\rangle, l} \mathbf{D}_{ij,l} \cdot (\mathbf{S}_{i,l} \times \mathbf{S}_{j,l}) + J_c \sum_{\langle ll' \rangle, i} \mathbf{S}_{i,l} \cdot \mathbf{S}_{i,l'}. \tag{1.3}$$

Here, $J > 0$ is the nearest-neighbor (NN) intralayer antiferromagnetic coupling, $\mathbf{D}_{ij,l}$ is the NNN DMI, and $J_c > 0$ is the NN interlayer antiferromagnetic coupling. The study of the low-energy excitations (magnons) of this system reveals a rich topological structure characterized by lines of band touching.

2 System Geometry: Lattice and Reciprocal Space

To understand the magnon band structure, it is essential to first define the geometry of the crystal lattice in both real space and reciprocal space.

2.1 Real-Space Lattice

The system consists of two-dimensional (2D) honeycomb lattices stacked congruently along the z-axis.

The 2D Honeycomb Lattice

Each layer is a honeycomb lattice, which is not a Bravais lattice. It is a triangular Bravais lattice with a two-atom basis (sublattices A and B). The primitive lattice vectors are:

$$\mathbf{a}_1 = (\sqrt{3}a, 0, 0) \tag{1.4}$$

$$\mathbf{a}_2 = \left(\frac{\sqrt{3}}{2}a, \frac{3}{2}a, 0\right) \tag{1.5}$$

where a is the nearest-neighbor distance.

The 3D Stacked Lattice

The 3D crystal is formed by stacking these honeycomb layers directly on top of each other, separated by a distance c. The third primitive lattice vector is:

$$\mathbf{a}_3 = (0, 0, c) \tag{1.6}$$

2.2 Finding the Reciprocal Space Basis

The reciprocal basis vectors $(\vec{b}_1, \vec{b}_2, \vec{b}_3)$ are derived from the real-space primitive vectors $(\vec{a}_1, \vec{a}_2, \vec{a}_3)$ using the standard definitions. The volume of the real-space primitive unit cell is $V = \vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3) = \frac{3\sqrt{3}}{2}a^2c$.

The calculation yields the final reciprocal basis vectors:

$$\vec{b}_1 = \frac{2\pi}{a}\left(\frac{1}{\sqrt{3}}, -\frac{1}{3}, 0\right) \tag{1.7}$$

$$\vec{b}_2 = \frac{2\pi}{a}\left(0, \frac{2}{3}, 0\right) \tag{1.8}$$

$$\vec{b}_3 = \frac{2\pi}{c}(0, 0, 1) \tag{1.9}$$

2.3 High-Symmetry Points in Reciprocal Space

High-symmetry points in the first Brillouin zone (a hexagonal prism) are defined by fractional coordinates in the basis of the reciprocal vectors. Their explicit Cartesian coordinates are:

- Γ **Point (Center):** $\Gamma = (0, 0, 0)$
- M **Point (Edge Midpoint):** $M = \frac{1}{2}\vec{b}_1 = \frac{\pi}{a}\left(\frac{1}{\sqrt{3}}, -\frac{1}{3}, 0\right)$
- K **Point (Basal Corner):** $K = \frac{1}{3}\vec{b}_1 + \frac{1}{3}\vec{b}_2 = \frac{2\pi}{3a}\left(\frac{1}{\sqrt{3}}, \frac{1}{3}, 0\right)$
- A **Point (Top Face Center):** $A = \frac{1}{2}\vec{b}_3 = \frac{\pi}{c}(0, 0, 1)$
- L **Point (Vertical Edge Midpoint):** $L = M + A = \left(\frac{\pi}{a\sqrt{3}}, -\frac{\pi}{3a}, \frac{\pi}{c}\right)$
- H **Point (Top Corner):** $H = K + A = \left(\frac{2\pi}{3a\sqrt{3}}, \frac{2\pi}{9a}, \frac{\pi}{c}\right)$

3 Symmetry Analysis

The topological protection of the magnon spectrum relies on the discrete symmetries of the Hamiltonian and its ground state.

3.1 The Origin of PT Symmetry in the Collinear Antiferromagnet

The system possesses PT symmetry because the combined operation of Time-Reversal ($\mathcal{T}$) and space-Inversion ($\mathcal{P}$) leaves the Hamiltonian and its ground state unchanged, even though neither operation is a symmetry on its own.

We consider the collinear Néel ground state: spins on sublattice A point up ($\mathbf{S}_A = S\hat{z}$) and spins on sublattice B point down ($\mathbf{S}_B = -S\hat{z}$).

- **Time-Reversal ($\mathcal{T}$):** This operation reverses all spins: $\mathcal{T} : \mathbf{S} \to -\mathbf{S}$. Applying this yields $\mathbf{S}_A \to -S\hat{z}$ and $\mathbf{S}_B \to +S\hat{z}$. This is a different magnetic state, so $\mathcal{T}$ **is not a symmetry**.
- **Space-Inversion ($\mathcal{P}$):** This operation swaps the two sublattices: $\mathcal{P}$: A $\leftrightarrow$ B. Applying this yields spins on A are now $-S\hat{z}$ and spins on B are $+S\hat{z}$. This is also a different state, so $\mathcal{P}$ **is not a symmetry**.

Action of the Combined Symmetry ($\mathcal{PT}$)

If we apply both operations in succession:

1. **Initial State:** $\mathbf{S}_A = +S\hat{z}$; $\mathbf{S}_B = -S\hat{z}$.
2. **Apply $\mathcal{T}$:** State becomes $\mathbf{S}_A = -S\hat{z}$; $\mathbf{S}_B = +S\hat{z}$.
3. **Apply $\mathcal{P}$:** The sublattices are swapped. The spins on A ($-S\hat{z}$) move to B. The spins on B ($+S\hat{z}$) move to A.
4. **Final State:** $\mathbf{S}_A = +S\hat{z}$; $\mathbf{S}_B = -S\hat{z}$.

The final state is identical to the initial state. Therefore, **PT is a symmetry** of the collinear antiferromagnet.

3.2 Symmetry Analysis: DMI vs. External Magnetic Field

We analyze how different terms transform under $\mathcal{T}$ and $\mathcal{P}$.

Why the DMI Term Preserves PT Symmetry

The Dzyaloshinskii-Moriya interaction is given by:

$$\mathcal{H}_{DM} = \sum_{\langle\langle ij \rangle\rangle} \mathbf{D}_{ij} \cdot (\mathbf{S}_i \times \mathbf{S}_j) \tag{1.10}$$

- Under $\mathcal{P}$: $\mathbf{D}_{ij}$ (a true vector) flips sign, while $(\mathbf{S}_i \times \mathbf{S}_j)$ (an axial vector) does not. Thus, $\mathcal{P}(\mathcal{H}_{DM}) = -\mathcal{H}_{DM}$. $\mathcal{P}$ **is broken**.
- Under $\mathcal{T}$: $\mathbf{D}_{ij}$ (from spin-orbit coupling) flips sign, and $\mathbf{S}$ flips sign. The cross product $(\mathbf{S}_i \times \mathbf{S}_j)$ is invariant. Thus, $\mathcal{T}(\mathcal{H}_{DM}) = -\mathcal{H}_{DM}$. $\mathcal{T}$ **is broken**.

Applying both operations gives:

$$\mathcal{PT}(\mathcal{H}_{DM}) = \mathcal{P}(\mathcal{T}(\mathcal{H}_{DM})) = \mathcal{P}(-\mathcal{H}_{DM}) = -(\mathcal{P}(\mathcal{H}_{DM})) = -(-\mathcal{H}_{DM}) = +\mathcal{H}_{DM} \quad (1.11)$$

The DM interaction is **invariant under the combined PT operation**. Because it respects the protecting symmetry of the DNL phase, it cannot open a gap in the magnon spectrum.

Why the External Magnetic Field Breaks PT Symmetry

The Zeeman term from an external magnetic field is given by:

$$\mathcal{H}_H = -H \sum_i S_i^z \quad (1.12)$$

- Under $\mathcal{P}$: S_i^z (axial vector z-component) is invariant. Thus, $\mathcal{P}(\mathcal{H}_H) = \mathcal{H}_H$. $\mathcal{P}$ **is preserved**.
- Under $\mathcal{T}$: S_i^z flips sign. Thus, $\mathcal{T}(\mathcal{H}_H) = -\mathcal{H}_H$. $\mathcal{T}$ **is broken**.

Applying both operations gives:

$$\mathcal{PT}(\mathcal{H}_H) = \mathcal{P}(\mathcal{T}(\mathcal{H}_H)) = \mathcal{P}(-\mathcal{H}_H) = -(\mathcal{P}(\mathcal{H}_H)) = -(\mathcal{H}_H) = -\mathcal{H}_H \quad (1.13)$$

The Zeeman term is **not invariant under the combined PT operation**. Because it explicitly breaks the symmetry that protects the DNLs, the external magnetic field acts as a perturbation that lifts the band degeneracy and opens the topological gap.

4 Hamiltonian Bosonization and Diagonalization

To analyze the magnon spectrum at low temperatures, we employ the Holstein-Primakoff (HP) transformation to express the spin operators in terms of bosonic creation and annihilation operators. This is followed by a Fourier transform and a Bogoliubov transformation to diagonalize the Hamiltonian.

4.1 Derivation of the Momentum-Space Hamiltonian

For the collinear Néel ordered state, we define separate transformations for the two sublattices A and B. For a site i on sublattice A and a site j on sublattice B:

$$S_{i,A}^z = S - a_i^\dagger a_i, \qquad S_{i,A}^+ \approx \sqrt{2S} a_i, \qquad S_{i,A}^- \approx \sqrt{2S} a_i^\dagger \quad (1.14)$$

$$S_{j,B}^z = -S + b_j^\dagger b_j, \qquad S_{j,B}^+ \approx \sqrt{2S} b_j^\dagger, \qquad S_{j,B}^- \approx \sqrt{2S} b_j \quad (1.15)$$

We then Fourier transform from real space operators (a_i, b_j) to momentum space operators $(a_{\mathbf{k}}, b_{\mathbf{k}})$.

After combining all terms, the final quadratic Hamiltonian is $\mathcal{H} = \frac{1}{2}\sum_{\mathbf{k}} \Psi^\dagger(\mathbf{k})\mathcal{H}(\mathbf{k})\Psi(\mathbf{k})$, with the basis $\Psi^\dagger(\mathbf{k}) = (a^\dagger_{\mathbf{k},A}, a^\dagger_{\mathbf{k},B}, a_{-\mathbf{k},A}, a_{-\mathbf{k},B})$ and the matrix:

$$\mathcal{H}(\mathbf{k}) = \begin{pmatrix} \mathcal{A}(\mathbf{k}) & \mathcal{B}(\mathbf{k}) \\ \mathcal{B}^*(-\mathbf{k}) & \mathcal{A}^*(-\mathbf{k}) \end{pmatrix}, \tag{1.16}$$

where $\mathcal{A}(\mathbf{k}) = [f_0 + f^D(\mathbf{k}_{||})]I_{2\times 2}$ and $\mathcal{B}(\mathbf{k}) = \begin{pmatrix} f(k_z) & f(\mathbf{k}_{||}) \\ f^*(\mathbf{k}_{||}) & f(k_z) \end{pmatrix}$. The functions are defined as:

- $f_0 = 3JS + 2J_cS$
- $f(k_z) = 2J_cS\cos(k_z)$
- $f(\mathbf{k}_{||}) = JS(1 + e^{-i\mathbf{k}_{||}\cdot\mathbf{a}_1} + e^{-i\mathbf{k}_{||}\cdot\mathbf{a}_2})$
- $f^D(\mathbf{k}_{||}) = 2DS(\sin(\mathbf{k}_{||}\cdot\mathbf{a}_1) - \sin(\mathbf{k}_{||}\cdot\mathbf{a}_2) + \sin(\mathbf{k}_{||}\cdot(\mathbf{a}_2 - \mathbf{a}_1)))$

4.2 Detailed Bogoliubov Diagonalization

The Hamiltonian is diagonalized via a Bogoliubov transformation, which is equivalent to solving the generalized eigenvalue problem $\mathcal{H}(\mathbf{k})\Phi = E\sigma_z\Phi$, where $\sigma_z = \text{diag}(1, 1, -1, -1)$. This leads to the characteristic equation:

$$\det\Big((\mathcal{A}(\mathbf{k})^2 - E^2 I) - \mathcal{B}(\mathbf{k})\mathcal{B}^\dagger(\mathbf{k})\Big) = 0. \tag{1.17}$$

The eigenvalues $\lambda_{\mathcal{B}}$ of $\mathcal{B}(\mathbf{k})$ are $\lambda_{\mathcal{B},\pm} = f(k_z) \pm |f(\mathbf{k}_{||})|$. Substituting this into the characteristic equation gives:

$$(f_0 + f^D(\mathbf{k}_{||}))^2 - E^2 = (f(k_z) \pm |f(\mathbf{k}_{||})|)^2. \tag{1.18}$$

Solving for E yields the final dispersion for the two magnon bands:

$$E_\pm(\mathbf{k}) = \sqrt{(f_0 + f^D(\mathbf{k}_{||}))^2 - (f(k_z) \pm |f(\mathbf{k}_{||})|)^2}. \tag{1.19}$$

5 The Dirac Nodal Line Phase

At zero magnetic field, the system exhibits a topological semi-metal phase. The bands become degenerate, $E_+(\mathbf{k}) = E_-(\mathbf{k})$, when the condition $f(k_z)\cdot|f(\mathbf{k}_{||})| = 0$ is met. This degeneracy does not occur at isolated points but along continuous lines in the Brillouin zone, known as **Dirac Nodal Lines (DNLs)**.

As shown in Figure 1.2, the bands touch along high-symmetry lines. The dashed red lines show that this touching persists even when the DMI ($D = 0.2$) is included, as the DMI preserves the protecting PT symmetry.

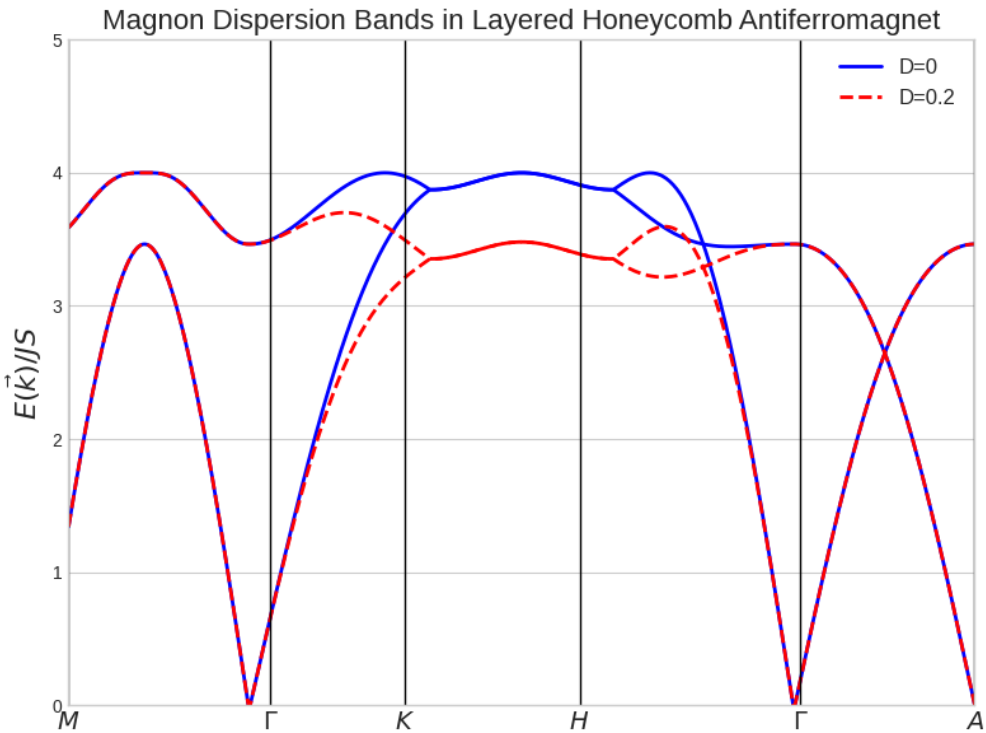

Figure 1.2: Magnon dispersion bands along high-symmetry lines in the Brillouin zone for $J_c/J = 0.5$. The solid blue lines correspond to $D = 0$, showing Dirac Nodal Lines (DNLs) where the bands touch. The dashed red lines ($D = 0.2$) show that the DNLs persist even with DMI.

6 Topology of the DNL Phase

The DNLs are not accidental degeneracies; they are topologically protected by the PT symmetry of the system.

6.1 PT Symmetry and its Role in DNL Topology

The invariance of the Hamiltonian under PT symmetry is what **protects** the Dirac Nodal Lines.

- **Protection Against Gapping:** For a gap to open and destroy a nodal line, the perturbation added to the system **must break PT symmetry**. Any perturbation that preserves PT symmetry (like the intralayer DMI in this model) is incapable of gapping the nodal lines.

- **Vanishing Berry Curvature:** PT symmetry imposes a powerful mathematical constraint on the Berry curvature, $\mathbf{\Omega}(\mathbf{k})$. Time-reversal requires $\mathbf{\Omega}(\mathbf{k}) = -\mathbf{\Omega}(-\mathbf{k})$, while inversion requires $\mathbf{\Omega}(\mathbf{k}) = \mathbf{\Omega}(-\mathbf{k})$. For a system that has both (i.e., is PT symmetric), the only way to satisfy both conditions is for the Berry curvature to be **identically**

zero everywhere:

$$\mathbf{\Omega}(\mathbf{k}) = 0 \tag{1.20}$$

6.2 Topological Invariant: The Berry Phase

For a gapless system like a DNL semi-metal, the primary topological invariant is the **Berry phase** (or Zak phase), γ_C, calculated along a closed loop C in momentum space. It is defined as the line integral of the Berry connection, $\mathbf{A}_n(\mathbf{k})$:

$$\gamma_C = \oint_C \mathbf{A}_n(\mathbf{k}) \cdot d\mathbf{k} \tag{1.21}$$

where $\mathbf{A}_n(\mathbf{k}) = i\langle u_n(\mathbf{k})|\nabla_{\mathbf{k}}|u_n(\mathbf{k})\rangle$.

If the loop C **does not** enclose a nodal line, the Berry phase is trivial, $\gamma_C = 0$. However, if the loop C **does** enclose a segment of a nodal line, the Berry phase is quantized to a non-trivial value of $\gamma_C = \pm\pi$. This quantized, non-zero value acts as a topological charge.

6.3 Berry Curvature and the Chern Number in the DNL Phase

The **Chern number**, $\mathcal{C}_n$, is a topological invariant for gapped 2D systems, calculated by integrating the Berry curvature over the 2D Brillouin zone. Since the Berry curvature is identically zero everywhere in the PT-symmetric DNL phase, $\mathbf{\Omega}_n(\mathbf{k}) = 0$, the Chern number is trivially zero. The Chern number is not the appropriate invariant to describe this gapless phase; the π Berry phase is.

6.4 Calculation of the Quantized π Berry Phase

Near a band-touching point (like K), the low-energy dynamics are described by an effective 2x2 Dirac Hamiltonian:

$$\mathcal{H}_{\text{eff}}(\mathbf{q}) = v(q_x\sigma_x + q_y\sigma_y) + \Delta\sigma_z = \begin{pmatrix} \Delta & v(q_x - iq_y) \\ v(q_x + iq_y) & -\Delta \end{pmatrix} \tag{1.22}$$

Here, $\mathbf{q} = \mathbf{k} - \mathbf{K}$ is the momentum deviation and Δ is a mass term. The DNL corresponds to $\Delta = 0$.

The Berry phase γ_C for a loop enclosing the DNL ($\mathbf{q} = 0$) can be found by integrating the Berry curvature Ω_z over the area S enclosed by the loop (via Stokes' theorem).

$$\gamma_C = \iint_S \Omega_z \, dq_x dq_y \tag{1.23}$$

The Berry curvature for the lower band of this effective model is:

$$\Omega_z = -\frac{v^2\Delta}{2(v^2q^2 + \Delta^2)^{3/2}} \tag{1.24}$$

Integrating this over a circular area of radius R yields:

$$\gamma_C = -\pi\left[1 - \frac{\Delta}{\sqrt{v^2R^2 + \Delta^2}}\right] \tag{1.25}$$

In the DNL phase, we take the gapless limit $\Delta \to 0$:

$$\gamma_C = \lim_{\Delta \to 0} \left(-\pi \left[1 - \frac{\Delta}{\sqrt{v^2 R^2 + \Delta^2}} \right] \right) = -\pi[1 - 0] = -\pi \tag{1.26}$$

This confirms the quantized π Berry phase, the topological invariant protecting the nodal line.

7 Absence of the Magnon Spin Nernst Effect

A key prediction is that the **magnon spin Nernst effect** is strictly absent in the DNL phase ($H = 0$).

7.1 The Magnon Spin Nernst Effect and Berry Curvature

The spin Nernst effect describes the generation of a transverse spin current J_y^s from a longitudinal temperature gradient $-\nabla_x T$. The spin Nernst conductivity α_{xy}^s is directly proportional to the integral of the Berry curvature:

$$\alpha_{xy}^s \propto \int_{BZ} d^3k \sum_n g(E_n(\mathbf{k}))\Omega_{xy}^n(\mathbf{k}) \tag{1.27}$$

If the Berry curvature $\Omega_{xy}^n(\mathbf{k})$ is identically zero, the spin Nernst conductivity must also be zero.

7.2 Mathematical Derivation of Vanishing Berry Curvature

As shown in the original work [12], the Berry curvature for this bosonic system can be calculated from the paraunitary matrix $\mathcal{U}_k$ that diagonalizes the Hamiltonian.

$$[\Omega_{\alpha\beta}(\mathbf{k})]_{nn} = -2\Im \left[\left(\tau_3 \frac{\partial \mathcal{U}_k^\dagger}{\partial k_\alpha} \right) \left(\tau_3 \frac{\partial \mathcal{U}_k}{\partial k_\beta} \right) \right]_{nn} \tag{1.28}$$

Due to the specific structure of $\mathcal{U}_k$ in the DNL phase (which is a direct result of PT symmetry), the diagonal elements of the product matrix $\left(\tau_3 \frac{\partial \mathcal{U}_k^\dagger}{\partial k_\alpha} \right) \left(\tau_3 \frac{\partial \mathcal{U}_k}{\partial k_\beta} \right)$ are **purely real**. Therefore, their imaginary part is zero:

$$\Im[\dots]_{nn} = 0 \tag{1.29}$$

This leads to the conclusion that the Berry curvature for every band is identically zero:

$$[\Omega_{\alpha\beta}(\mathbf{k})]_{nn} = 0 \tag{1.30}$$

7.3 Physical Interpretation: The Role of PT Symmetry

The mathematical result that $\Omega(\mathbf{k}) = 0$ is a direct consequence of the underlying **PT symmetry** of the DNL phase. Since the Berry curvature is zero, the integral for the spin Nernst conductivity is also zero, and thus the **magnon spin Nernst effect is absent** in this system.

8 Analysis of NNN Interlayer Spin Canting Interaction

In this section, we extend the model to include a next-nearest-neighbor (NNN) Dzyaloshinskii-Moriya-like interaction between the layers. We will show that this interaction breaks the protecting PT symmetry and destroys the DNL phase.

8.1 Spin Hamiltonian with NNN Interlayer Spin Canting Interaction

We add a new term, $\mathcal{H}_{D_{c,NNN}}$, to the Hamiltonian:

$$\mathcal{H} = \mathcal{H}_J + \mathcal{H}_D + \mathcal{H}_{J_c} + \mathcal{H}_{D_{c,NNN}} \tag{1.31}$$

where the new interaction is:

$$\mathcal{H}_{D_{c,NNN}} = \sum_{\langle i,j\rangle,\langle l,l'\rangle} \mathbf{D}_c \cdot (\mathbf{S}_{i,l} \times \mathbf{S}_{j,l'}) \tag{1.32}$$

This term describes an interaction between a spin on site i (sublattice A) in layer l and a neighboring spin on site j (sublattice B) in an adjacent layer l'. We assume $\mathbf{D}_c = D_c\hat{z}$.

8.2 Action of the NNN Interlayer Spin Canting Interaction

This NNN interlayer DMI-like term breaks the protecting PT symmetry of the original Hamiltonian. While the intralayer DMI ($\mathcal{H}_D$) was invariant under the combined PT operation, this new term is not, as it couples layers in a way that breaks the simple inversion symmetry between them.

8.3 Bosonization and Momentum Space Representation

We bosonize this new term and add it to the Hamiltonian matrix. The basis remains $\Psi^\dagger(\mathbf{k}) = (a^\dagger_{\mathbf{k}}, b^\dagger_{\mathbf{k}}, a_{-\mathbf{k}}, b_{-\mathbf{k}})$. The new term $\mathcal{H}_{D_{c,NNN}}$ introduces new off-diagonal elements into the $\mathcal{A}(\mathbf{k})$ block, which describes single-magnon hopping. The blocks become:

$$\mathcal{A}(\mathbf{k}) = \begin{pmatrix} f_0 + f^D(\mathbf{k}_{||}) & if^{D_c}(\mathbf{k}) \\ -if^{D_c*}(\mathbf{k}) & f_0 - f^D(\mathbf{k}_{||}) \end{pmatrix} \tag{1.33}$$

$$\mathcal{B}(\mathbf{k}) = \begin{pmatrix} f(\mathbf{k}_z) & f(\mathbf{k}_{||}) \\ f^*(\mathbf{k}_{||}) & f(\mathbf{k}_z) \end{pmatrix} \tag{1.34}$$

The new function $f^{D_c}(\mathbf{k})$ arises from the NNN interlayer spin canting interaction.

8.4 Detailed Derivation of the Hamiltonian Matrix Elements

This section provides a step-by-step derivation of the functions that constitute the momentum-space Hamiltonian matrix, $\mathcal{H}(\mathbf{k})$.

Derivation of f_0 (On-site Energy Term)

The on-site energy f_0 originates from the $S^z S^z$ parts of the Heisenberg exchange.

- **Intralayer ($\mathcal{H}_J$):** $J \sum S^z_{i,l} S^z_{j,l} \approx \sum 3JSa^\dagger_{i,l} a_{i,l}$.
- **Interlayer ($\mathcal{H}_{J_c}$):** $J_c \sum S^z_{i,l} S^z_{i,l'} \approx \sum 2J_c S a^\dagger_{i,l} a_{i,l}$.

Combining these gives the total on-site energy:

$$\mathbf{f_0 = 3JS + 2J_cS} \tag{1.35}$$

Derivation of $f(\mathbf{k}_{||})$ (Intralayer Heisenberg Pairing Term)

This function arises from the in-plane components of $\mathcal{H}_J$, $\frac{J}{2}(S^+_{i,l} S^-_{j,l} + S^-_{i,l} S^+_{j,l}) \approx JS(a_{i,l} b_{j,l} + a^\dagger_{i,l} b^\dagger_{j,l})$. After Fourier transform, this pairing term becomes:

$$\mathbf{f(k_{||}) = JS(1 + e^{-ik_{||}\cdot a_1} + e^{-ik_{||}\cdot a_2})} \tag{1.36}$$

Derivation of $f(k_z)$ (Interlayer Heisenberg Pairing Term)

This function comes from the in-plane components of $\mathcal{H}_{J_c}$, which couple same-sublattice sites in adjacent layers: $\frac{J_c}{2}(S^+_{i,l} S^-_{i,l'} + S^-_{i,l} S^+_{i,l'}) \approx J_c S(a_{i,l} a_{i,l'} + a^\dagger_{i,l} a^\dagger_{i,l'})$. The Fourier transform yields:

$$\mathbf{f(k_z) = 2J_cS}\cos(\mathbf{k_z}) \tag{1.37}$$

Derivation of $f^D(\mathbf{k}_{||})$ (Intralayer DMI Term)

This function arises from the intralayer NNN DMI, $\mathcal{H}_D \approx \sum -i\nu_{ij} DS(a_i a^\dagger_j - a^\dagger_i a_j)$. This is a hopping term on the same sublattice. The Fourier transform gives:

$$\mathbf{f^D(k_{||}) = 2DS}(\sin(\mathbf{k_{||}\cdot a_1}) - \sin(\mathbf{k_{||}\cdot a_2}) + \sin(\mathbf{k_{||}\cdot (a_2 - a_1)})) \tag{1.38}$$

Derivation of $f^{D_c}(\mathbf{k})$ (NNN Interlayer Spin Canting Term)

This new function describes hopping between different sublattices on adjacent layers. It comes from $\mathcal{H}_{D_c} = \sum D_c(\mathbf{S}_{i,l} \times \mathbf{S}_{j,l'})_z$.

$$D_c(S^x_{i,l} S^y_{j,l'} - S^y_{i,l} S^x_{j,l'}) \approx -iD_c S(a_{i,l} b^\dagger_{j,l'} - a^\dagger_{i,l} b_{j,l'})$$

This is a hopping term that modifies the off-diagonal elements of the $\mathcal{A}(\mathbf{k})$ block. The Fourier transform of $\sum iD_c S a^\dagger_{i,l} b_{j,l'}$ over NN bonds $\boldsymbol{\delta}$ and adjacent layers ($l' = l \pm 1$) gives:

$$\sum_{\mathbf{k}} iD_c S \left(\sum_{\delta} e^{-i\mathbf{k}_{||}\cdot\delta} \right) (e^{ik_z} + e^{-ik_z}) a^\dagger_{\mathbf{k}} b_{\mathbf{k}} = i \cdot 2D_c S \cos(k_z) \frac{f^*(\mathbf{k}_{||})}{JS} a^\dagger_{\mathbf{k}} b_{\mathbf{k}}$$

We define the magnitude of this coefficient as:

$$\mathbf{f^{D_c}(k) = 2D_cS}\cos(\mathbf{k_z})\frac{|\mathbf{f(k_{||})}|}{\mathbf{JS}} \tag{1.39}$$

8.5 Loss of the DNL Phase due to Interlayer Spin-Canting Interaction

The new off-diagonal term $f^{D_c}(\mathbf{k})$ in the $\mathcal{A}(\mathbf{k})$ block explicitly breaks the PT symmetry that protected the DNLs. When the full Hamiltonian is diagonalized, this term lifts the degeneracy between the magnon bands.

The plot in Figure 1.3 illustrates the consequences of this analytical result. The green dotted lines, representing the model with the NNN interlayer spin-canting interaction (D_c), show a **clear gap opening** along paths like K-H. This confirms that the interaction breaks the protecting PT symmetry and **destroys the Dirac Nodal Line phase** present in the original models.

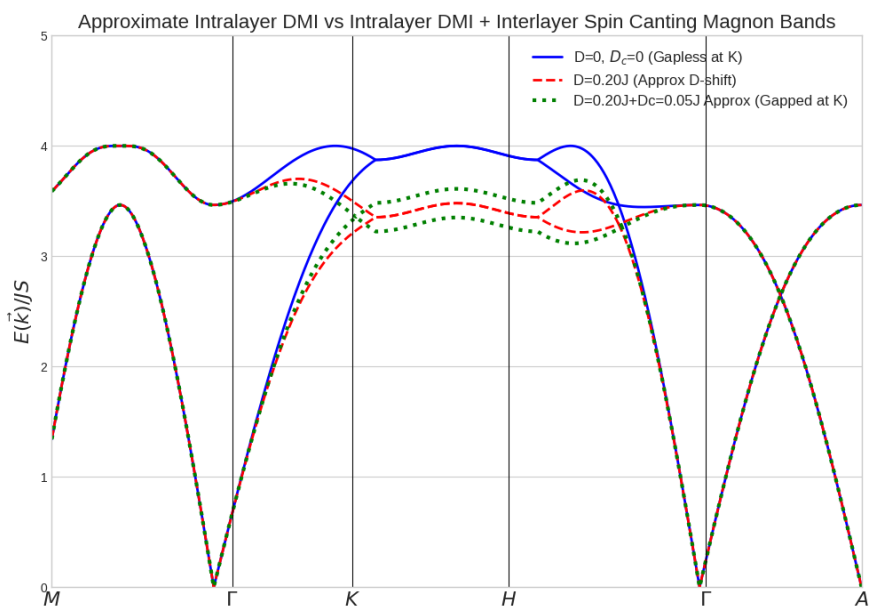

Figure 1.3: Magnon dispersion bands comparing the original model (blue and red lines) with the extended model including the NNN interlayer DMI (green dotted lines). The new interaction opens a gap between the magnon bands, indicating the loss of the DNL phase.

8.6 Lattice Visualization

Figure 1.4 provides a 3D visualization of the stacked honeycomb lattice, showing the spin configuration and the various interaction paths.

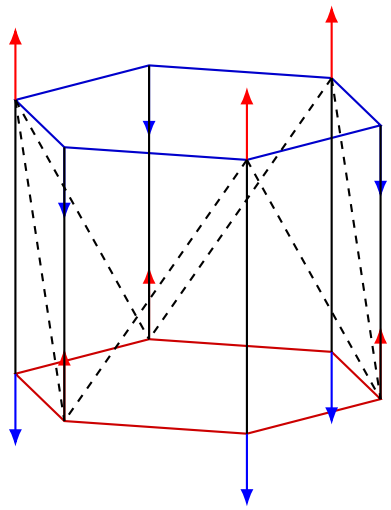

Figure 1.4: A 3D visualization of the stacked honeycomb lattice with a collinear Néel antiferromagnetic (AFM) spin configuration. The solid vertical lines represent the nearest-neighbor interlayer coupling (J_c), while the dashed lines indicate the next-nearest-neighbor interlayer spin canting interaction paths (D_c).

9 Conclusion and Further Discussion

In summary, our study confirms that the Dirac Nodal Lines in layered honeycomb antiferromagnets are topologically protected by PT symmetry. This protection guarantees the band-touching and leads to a vanishing Berry curvature, which in turn forbids the magnon spin Nernst effect.

Furthermore, we extended the model to include a next-nearest-neighbor interlayer spin-canting interaction. We analytically derived the new Hamiltonian and demonstrated that this interaction **breaks the protecting PT symmetry**. As confirmed by numerical calculations (Fig. 1.3), this symmetry-breaking term lifts the band degeneracy and **opens a gap, destroying the topological DNL phase**. These results provide deeper insight into the interplay between symmetry, topology, and magnon band structure, and suggest possible routes for engineering topological magnonic phases through careful control of interlayer couplings.

CHAPTER 2

CONCLUDING REMARKS

In this report we have studied the role of inter-layer spin canting in layered antiferromagnetic systems. We began by exploring the fundamental concepts of topological magnonics, including the significance of Berry curvature and Chern numbers in characterizing topological phases. We then delved into the specific case of layered antiferromagnets, examining how inter-layer interactions and spin canting influence the magnon band structure and topological properties. Our analysis revealed that inter-layer spin canting can induce non-trivial topological phases, leading to loss of the DNL phase as predicted in the paper studied [12].
I look forward to exploring these phenomena further and I am curious to see experimental perspectives on this research. I would like to work with more involved systems like Altermagnetic materials and see how exotic phenomena and/or topological superconductivity can arise from them. [14]

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